HAZOUME Exaucé SISR B

Compte rendu de projet : Mise en place de deux serveurs Web en haute disponibilité avec Heartbeat

1. Introduction

L'objectif de ce projet est de mettre en place deux serveurs Web sous Apache2, configurés avec des pages personnalisées, et d'assurer leur haute disponibilité grâce au service **Heartbeat**. Un client Windows est utilisé pour tester l'accessibilité et la bascule automatique entre les deux serveurs.

2. Configuration des machines

- **srvWeb1** :
  - Hostname : srvWeb1
  - Adresse IP : `172.20.0.11/24`
  - Interface réseau : `ens33`

- **srvWeb2** :
  - Hostname : srvWeb1
  - Adresse IP : `172.20.0.12/24`
  - Interface réseau : `ens33`

- **Client Windows** :
  - Adresse IP : `172.20.0.100/24`
  - Interface réseau : LAN Segment

3. Personnalisation des pages Apache2

1. Suppression de la page par défaut d'Apache.
2. Téléchargement et décompression du template **The Grill**.
3. Modification du fichier de configuration Apache pour pointer vers le dossier `thegrill-master`.

Tout cela donc fait, procédons aux vérification

Validations :

- srvWeb1 : hostname -> srvWeb1 :

Vous le remarquerez surement j'ai changé le nom des machines pas au tout début mais lors des configuration, c'est pour ça au tout début c'est root_debian et ensuite root srvwb1 ou 2

```
root@debian:/home/user# hostnamectl
 Static hostname: srvWeb1
       Icon name: computer-vm
         Chassis: vm
      Machine ID: f8b300b9681541eb9296981c596ee00b
         Boot ID: 6ba884ffed4349cca9142c0b8c1c7274
  Virtualization: vmware
Operating System: Debian GNU/Linux 12 (bookworm)
          Kernel: Linux 6.1.0-40-amd64
    Architecture: x86-64
 Hardware Vendor: VMware, Inc.
  Hardware Model: VMware Virtual Platform
Firmware Version: 6.00
root@debian:/home/user#
```

- srvWeb1 : ip a -> 172.20.0.11 :

```
user@debian: ~
user@debian:~$ su
Mot de passe :
root@debian:/home/user# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
qlen 1000
    link/ether 00:0c:29:04:26:05 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 172.20.0.11/24 brd 172.20.0.255 scope global ens33
       valid_lft forever preferred_lft forever
```

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- srvWeb1 : service apache2 status -> active running :

```
user@debian: ~
root@debian:/home/user# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qle
n 1000
    link/ether 00:0c:29:04:26:05 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 172.20.0.11/24 brd 172.20.0.255 scope global ens33
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe04:2605/64 scope link
       valid_lft forever preferred_lft forever
root@debian:/home/user# sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Mon 2025-11-10 23:49:21 CET; 11min ago
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 977 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 1008 (apache2)
      Tasks: 55 (limit: 2236)
     Memory: 15.4M
        CPU: 288ms
     CGroup: /system.slice/apache2.service
             ├─1008 /usr/sbin/apache2 -k start
             ├─1009 /usr/sbin/apache2 -k start
             └─1010 /usr/sbin/apache2 -k start

nov. 10 23:49:21 debian systemd[1]: Starting apache2.service - The Apache HTTP Server...
nov. 10 23:49:21 debian apachectl[1003]: AH00558: apache2: Could not reliably determine the s>
nov. 10 23:49:21 debian systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-17/17 (END)...skipping...
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Mon 2025-11-10 23:49:21 CET; 11min ago
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 977 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 1008 (apache2)
```

- srvWeb2 : hostname -> srvWeb2 :

```
root@debian:/home/user# hostnamectl
 Static hostname: srvWeb2
       Icon name: computer-vm
         Chassis: vm
      Machine ID: f8b300b9681541eb9296981c596ee00b
         Boot ID: 47161571fb6c425096a1c3b2db9738f2
  Virtualization: vmware
Operating System: Debian GNU/Linux 12 (bookworm)
          Kernel: Linux 6.1.0-40-amd64
    Architecture: x86-64
 Hardware Vendor: VMware, Inc.
  Hardware Model: VMware Virtual Platform
Firmware Version: 6.00
root@debian:/home/user#
```

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- srvWeb2 : ip a -> 172.20.0.12 :

```
user@srvWeb2: ~
user@srvWeb2:~$ su -
Mot de passe :
root@srvWeb2:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:85:1d:47 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 172.20.0.12/24 brd 172.20.0.255 scope global ens33
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe85:1d47/64 scope link
       valid_lft forever preferred_lft forever
root@srvWeb2:~#
```

- srvWeb2 : service apache2 status -> active running :

```
user@debian: ~
root@debian:/home/user# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:85:1d:47 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 172.20.0.12/24 brd 172.20.0.255 scope global ens33
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe85:1d47/64 scope link
       valid_lft forever preferred_lft forever
root@debian:/home/user# sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Mon 2025-11-10 23:48:58 CET; 15min ago
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 982 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 1014 (apache2)
      Tasks: 55 (limit: 2236)
     Memory: 17.4M
        CPU: 291ms
     CGroup: /system.slice/apache2.service
             ├─1014 /usr/sbin/apache2 -k start
             ├─1016 /usr/sbin/apache2 -k start
             └─1018 /usr/sbin/apache2 -k start

nov. 10 23:48:58 debian systemd[1]: Starting apache2.service - The Apache HTTP Server...
nov. 10 23:48:58 debian apachectl[1009]: AH00558: apache2: Could not reliably determine the server'>
nov. 10 23:48:58 debian systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-17/17 (END)...skipping...
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Mon 2025-11-10 23:48:58 CET; 15min ago
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 982 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
```

- srvWeb2 : ping 172.20.0.11 -> OK :

```
user@debian: ~
user@debian:~$ su
Mot de passe :
root@debian:/home/user# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
qlen 1000
    link/ether 00:0c:29:85:1d:47 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 172.20.0.12/24 brd 172.20.0.255 scope global ens33
       valid_lft forever preferred_lft forever
root@debian:/home/user# ping 172.20.0.11
PING 172.20.0.11 (172.20.0.11) 56(84) bytes of data.
64 bytes from 172.20.0.11: icmp_seq=1 ttl=64 time=1.42 ms
64 bytes from 172.20.0.11: icmp_seq=2 ttl=64 time=1.65 ms
64 bytes from 172.20.0.11: icmp_seq=3 ttl=64 time=1.69 ms
64 bytes from 172.20.0.11: icmp_seq=4 ttl=64 time=1.08 ms
64 bytes from 172.20.0.11: icmp_seq=5 ttl=64 time=1.59 ms
64 bytes from 172.20.0.11: icmp_seq=6 ttl=64 time=1.58 ms
^C
--- 172.20.0.11 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5006ms
rtt min/avg/max/mdev = 1.080/1.501/1.686/0.206 ms
root@debian:/home/user#
```

- Client Windows : ipconfig -> 172.20.0.100 :

```
Administrateur : C:\Windows\system32\cmd.exe
C:\Windows\system32>ipconfig

Configuration IP de Windows


Carte Ethernet Ethernet0 :

   Suffixe DNS propre à la connexion. . . :
   Adresse IPv6 de liaison locale. . . . .: fe80::41d5:22b7:3737:5c2b%5
   Adresse IPv4. . . . . . . . . . . . . .: 172.20.0.100
   Masque de sous-réseau. . . . . . . . . : 255.255.255.0
   Passerelle par défaut. . . . . . . . . : 172.20.0.1
```

- Client Windows : ping 172.20.0.11 -> OK :
- Client Windows : ping 172.20.0.12 -> OK :

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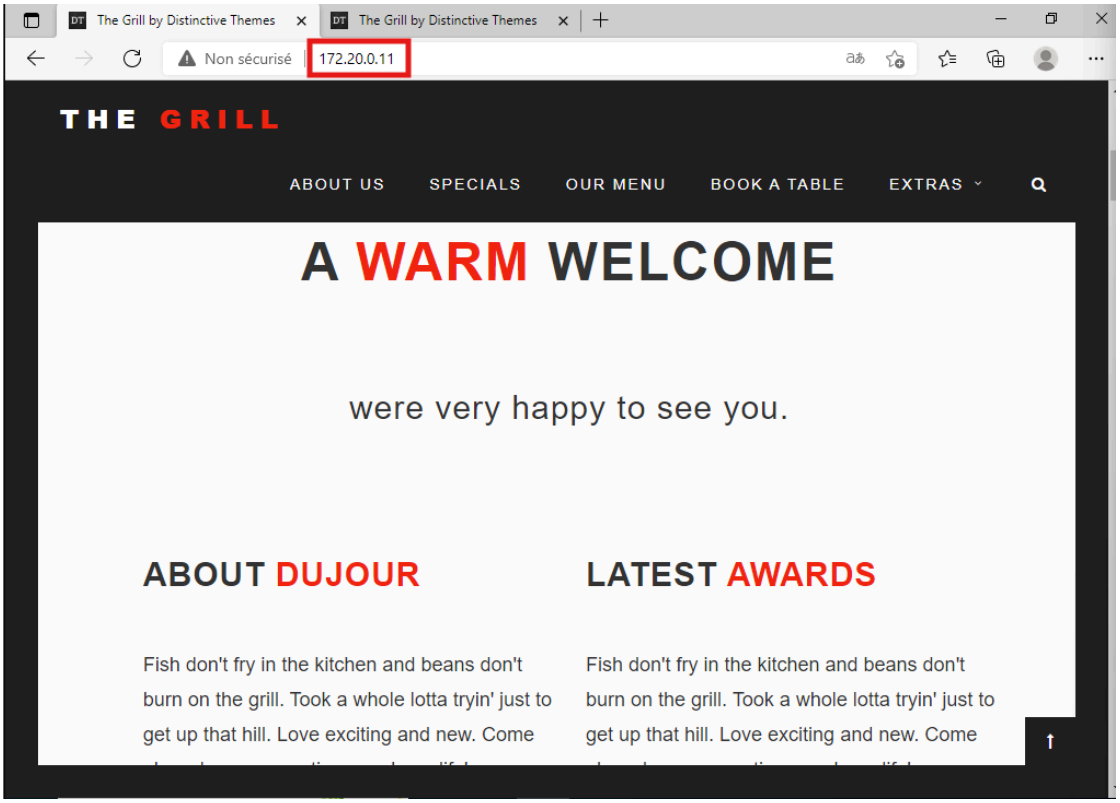
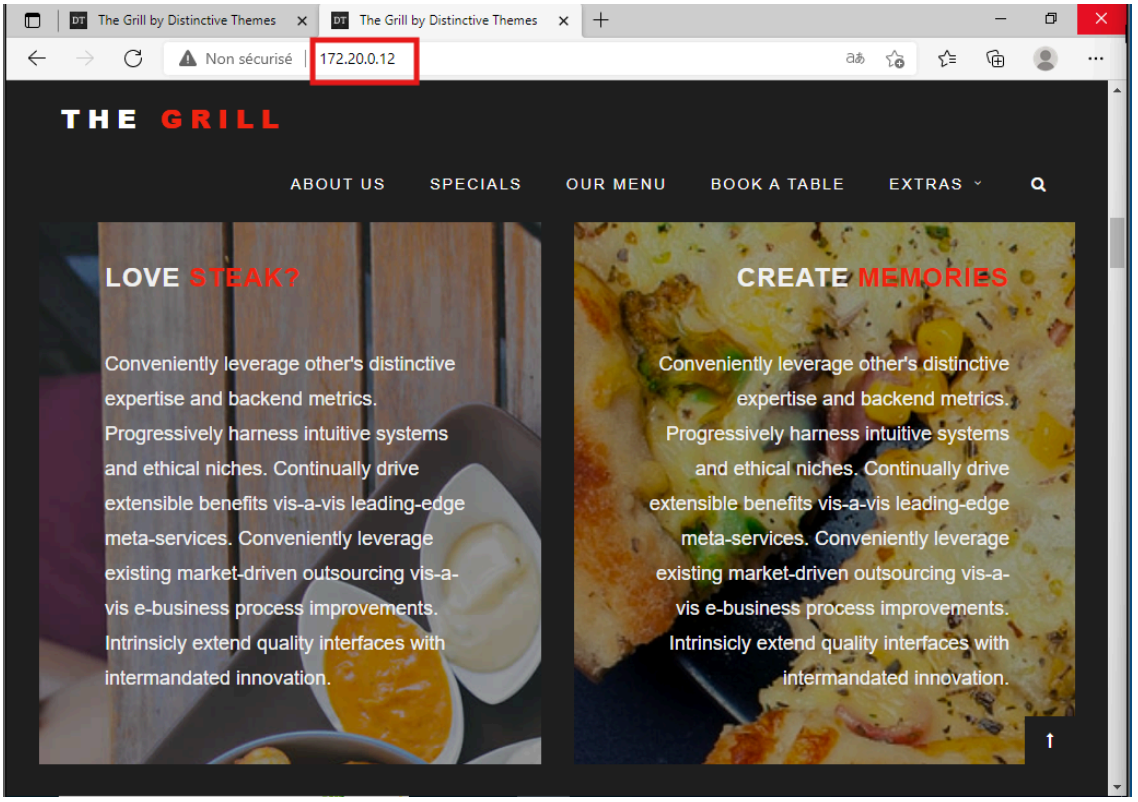
```
Administrateur : C:\Windows\system32\cmd.exe

C:\Windows\system32>ipconfig

Configuration IP de Windows


Carte Ethernet Ethernet0 :

   Suffixe DNS propre à la connexion. . . :
   Adresse IPv6 de liaison locale. . . . .: fe80::41d5:22b7:3737:5c2b%5
   Adresse IPv4. . . . . . . . . . . . . .: 172.20.0.100
   Masque de sous-réseau. . . . . . . . . : 255.255.255.0
   Passerelle par défaut. . . . . . . . . : 172.20.0.1

C:\Windows\system32>ping 172.20.0.11

Envoi d'une requête 'Ping'  172.20.0.11 avec 32 octets de données :
Réponse de 172.20.0.11 : octets=32 temps=1 ms TTL=64
Réponse de 172.20.0.11 : octets=32 temps<1ms TTL=64
Réponse de 172.20.0.11 : octets=32 temps=1 ms TTL=64
Réponse de 172.20.0.11 : octets=32 temps=1 ms TTL=64

Statistiques Ping pour 172.20.0.11:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 0ms, Maximum = 1ms, Moyenne = 0ms

C:\Windows\system32>ping 172.20.0.12

Envoi d'une requête 'Ping'  172.20.0.12 avec 32 octets de données :
Réponse de 172.20.0.12 : octets=32 temps=1 ms TTL=64
Réponse de 172.20.0.12 : octets=32 temps=1 ms TTL=64
Réponse de 172.20.0.12 : octets=32 temps=3 ms TTL=64
Réponse de 172.20.0.12 : octets=32 temps=1 ms TTL=64

Statistiques Ping pour 172.20.0.12:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 1ms, Maximum = 3ms, Moyenne = 1ms

C:\Windows\system32>
```

- Navigateur :
  - 172.20.0.11 -> The Grill 1 :



- 172.20.0.12 -> The Grill 2 :



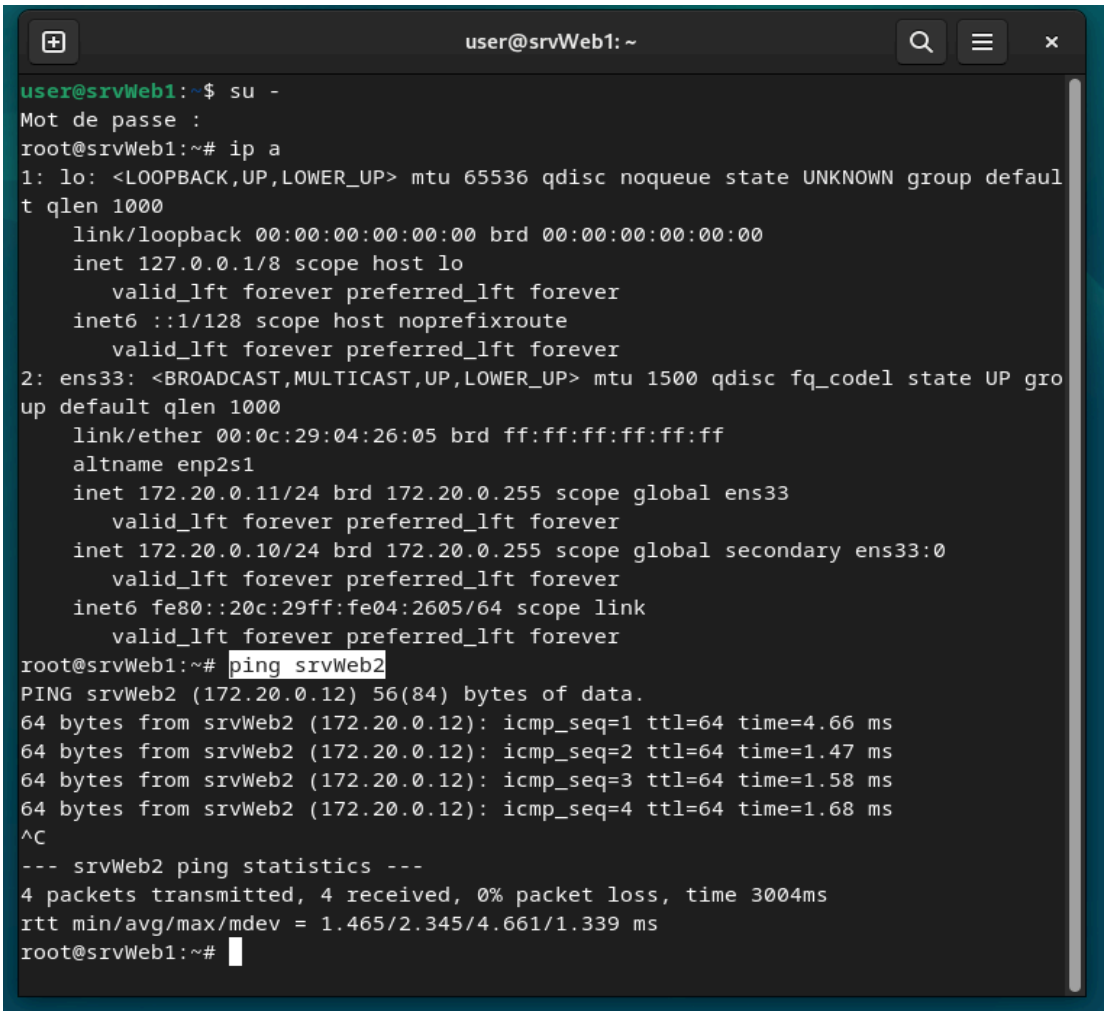
4. Configuration des fichiers hosts

Ajout des entrées suivantes dans `/etc/hosts` sur les deux serveurs :

Code
127.0.0.1 localhost
172.20.0.11 srvWeb1
172.20.0.12 srvWeb2

Validations :

- ping srvWeb1 -> OK :



- ping srvWeb2 -> OK :

```
user@srvWeb2: ~
user@srvWeb2:~$ su -
Mot de passe :
root@srvWeb2:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:85:1d:47 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 172.20.0.12/24 brd 172.20.0.255 scope global ens33
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe85:1d47/64 scope link
       valid_lft forever preferred_lft forever
root@srvWeb2:~# ping srvWeb1
PING srvWeb1 (172.20.0.11) 56(84) bytes of data.
64 bytes from srvWeb1 (172.20.0.11): icmp_seq=1 ttl=64 time=2.39 ms
64 bytes from srvWeb1 (172.20.0.11): icmp_seq=2 ttl=64 time=1.61 ms
64 bytes from srvWeb1 (172.20.0.11): icmp_seq=3 ttl=64 time=0.956 ms
64 bytes from srvWeb1 (172.20.0.11): icmp_seq=4 ttl=64 time=1.54 ms
64 bytes from srvWeb1 (172.20.0.11): icmp_seq=5 ttl=64 time=1.51 ms
^C
--- srvWeb1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 0.956/1.601/2.394/0.459 ms
root@srvWeb2:~#
```

5. Mise en place de Heartbeat

1. Vérification du lien symbolique `/etc/heartbeat` pointant vers `/etc/ha.d`.
2. Création des fichiers de configuration :

- **ha.cf** : configuration principale (logs, intervalle, deadtime, interface, nœuds).
- **haresources** : définition de l'IP virtuelle `172.20.0.10` et du service Apache2.
- **authkeys** : configuration de l'authentification (sécurisé avec `chmod 600`).

3. Désactivation du démarrage automatique d'Apache (géré par Heartbeat).
4. Arrêt des services Apache et Heartbeat pour préparer la configuration.

6. Validation du basculement

1. Démarrage de Heartbeat sur **srvWeb1** puis **srvWeb2**.

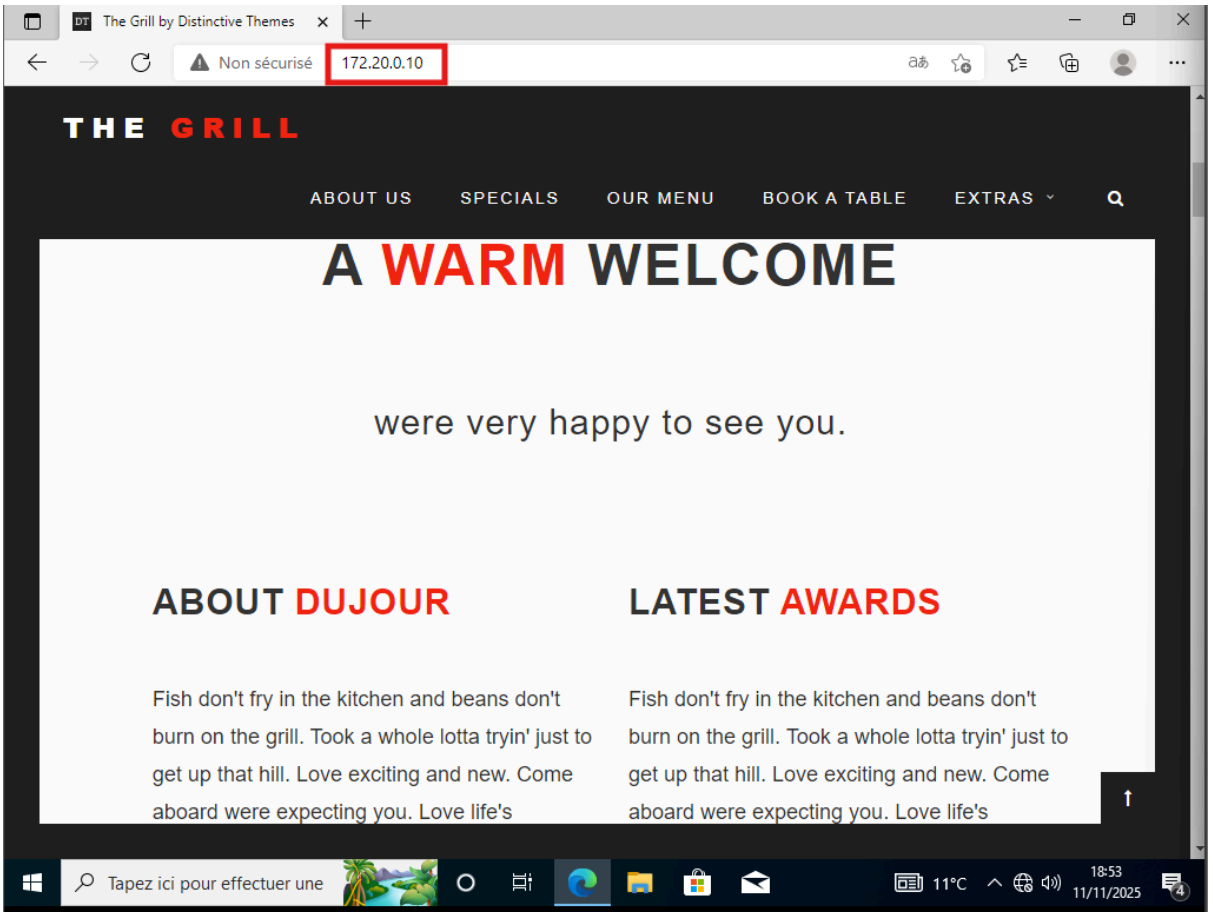
```
user@srvWeb1: ~
64 bytes from srvWeb2 (172.20.0.12): icmp_seq=2 ttl=64 time=1.47 ms
64 bytes from srvWeb2 (172.20.0.12): icmp_seq=3 ttl=64 time=1.58 ms
64 bytes from srvWeb2 (172.20.0.12): icmp_seq=4 ttl=64 time=1.68 ms
^C
--- srvWeb2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.465/2.345/4.661/1.339 ms
root@srvWeb1:~# sudo systemctl status heartbeat
● heartbeat.service - Heartbeat High Availability Cluster Communication and Mem>
     Loaded: loaded (/lib/systemd/system/heartbeat.service; enabled; preset: en>
     Active: active (running) since Tue 2025-11-18 14:13:37 CET; 42min ago
       Docs: man:heartbeat(8)
             http://www.linux-ha.org/wiki/Documentation
   Main PID: 1110 (heartbeat)
      Tasks: 4 (limit: 2236)
     Memory: 9.5M
        CPU: 7.903s
     CGroup: /system.slice/heartbeat.service
             ├─1110 "heartbeat: master control process"
             ├─1123 "heartbeat: FIFO reader"
             ├─1124 "heartbeat: write: bcast ens33"
             └─1125 "heartbeat: read: bcast ens33"

nov. 18 14:13:43 srvWeb1 ResourceManager(default)[1324]: info: Acquiring resour>
nov. 18 14:13:44 srvWeb1 /usr/lib/ocf/resource.d//heartbeat/IPaddr(IPaddr_172.2>
nov. 18 14:13:44 srvWeb1 IPaddr(IPaddr_172.20.0.10)[1485]: INFO: Using calculat>
nov. 18 14:13:44 srvWeb1 IPaddr(IPaddr_172.20.0.10)[1521]: INFO: eval ifconfig >
nov. 18 14:13:44 srvWeb1 /usr/lib/ocf/resource.d//heartbeat/IPaddr(IPaddr_172.2>
nov. 18 14:13:44 srvWeb1 ResourceManager(default)[1579]: info: Running /etc/ini>
nov. 18 14:13:45 srvWeb1 heartbeat[1298]: [1298]: info: local HA resource acqui>
nov. 18 14:13:45 srvWeb1 heartbeat[1110]: [1110]: info: Standby resource acquis>
nov. 18 14:13:45 srvWeb1 heartbeat[1110]: [1110]: info: Initial resource acquis>
nov. 18 14:13:45 srvWeb1 heartbeat[1110]: [1110]: info: remote resource transit>
lines 1-25/25 (END)
```

```
user@srvWeb2: ~
user@srvWeb2:~$ su -
Mot de passe :
root@srvWeb2:~# sudo systemctl status heartbeat
● heartbeat.service - Heartbeat High Availability Cluster Communication and Membership
     Loaded: loaded (/lib/systemd/system/heartbeat.service; enabled; preset: enabled)
     Active: active (running) since Tue 2025-11-18 14:09:06 CET; 50min ago
       Docs: man:heartbeat(8)
             http://www.linux-ha.org/wiki/Documentation
   Main PID: 1111 (heartbeat)
      Tasks: 4 (limit: 2236)
     Memory: 11.0M
        CPU: 9.489s
     CGroup: /system.slice/heartbeat.service
             ├─1111 "heartbeat: master control process"
             ├─1131 "heartbeat: FIFO reader"
             ├─1132 "heartbeat: write: bcast ens33"
             └─1133 "heartbeat: read: bcast ens33"

nov. 18 14:13:38 srvWeb2 heartbeat[1111]: [1111]: info: remote resource transition completed.
nov. 18 14:13:38 srvWeb2 heartbeat[1111]: [1111]: info: srvweb2 wants to go standby [foreign]
nov. 18 14:13:39 srvWeb2 heartbeat[1111]: [1111]: info: standby: srvweb1 can take our foreign resources
nov. 18 14:13:39 srvWeb2 heartbeat[2969]: [2969]: info: give up foreign HA resources (standby).
nov. 18 14:13:39 srvWeb2 heartbeat[2969]: [2969]: info: foreign HA resource release completed (standby).
nov. 18 14:13:39 srvWeb2 heartbeat[1111]: [1111]: info: Local standby process completed [foreign].
nov. 18 14:13:42 srvWeb2 heartbeat[1111]: [1111]: WARN: 1 lost packet(s) for [srvweb1] [11:13]
nov. 18 14:13:42 srvWeb2 heartbeat[1111]: [1111]: info: remote resource transition completed.
nov. 18 14:13:42 srvWeb2 heartbeat[1111]: [1111]: info: No pkts missing from srvweb1!
nov. 18 14:13:42 srvWeb2 heartbeat[1111]: [1111]: info: Other node completed standby takeover of foreign resources.
root@srvWeb2:~#
```

2. Vérification de l'IP virtuelle `172.20.0.10` sur **srvWeb1**.

```
root@debian:/home/user# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:04:26:05 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 172.20.0.11/24 brd 172.20.0.255 scope global ens33
       valid_lft forever preferred_lft forever
    inet 172.20.0.10/24 brd 172.20.0.255 scope global secondary ens33:0
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe04:2605/64 scope link
       valid_lft forever preferred_lft forever
root@debian:/home/user#
```

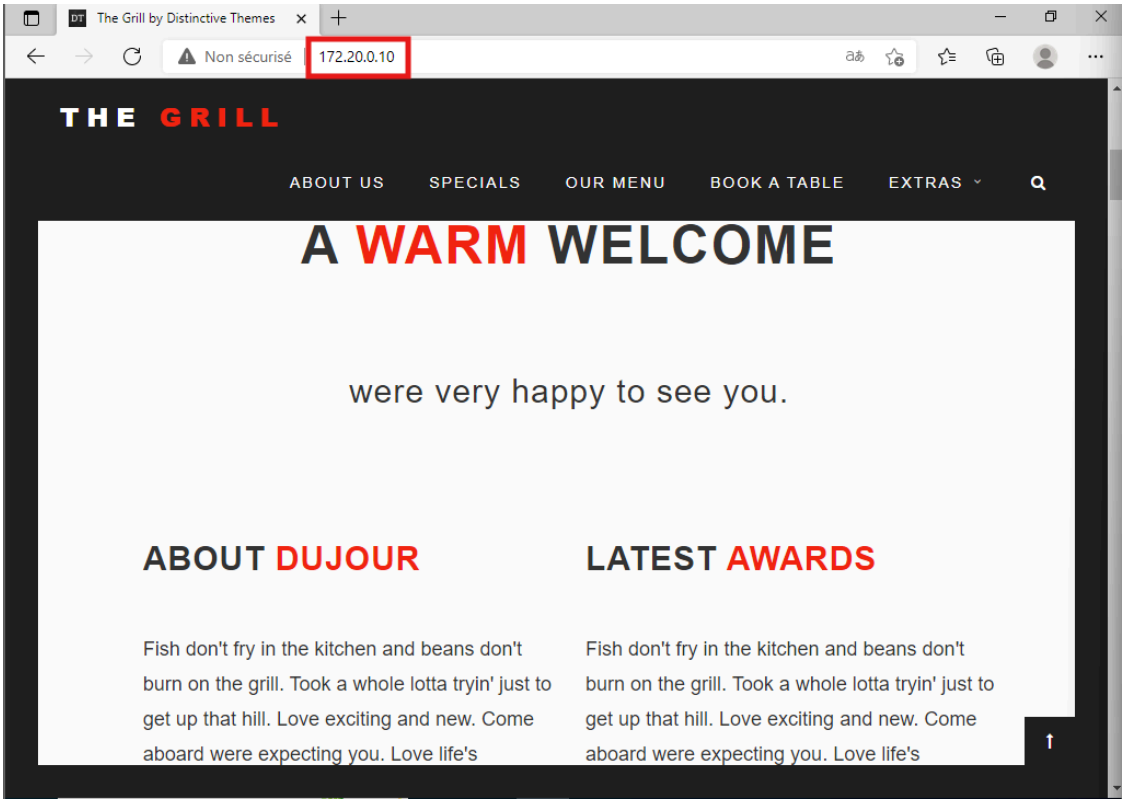
3. Depuis le client Windows : accès à `http://172.20.0.10` -> affichage de **The Grill 1** :



4. Simulation de panne : déconnexion de l'interface réseau de **srvWeb1**
5. Vérification sur **srvWeb2** : apparition de l'IP virtuelle `172.20.0.10`.

```
user@debian: ~
 Static hostname: srvWeb2
       Icon name: computer-vm
         Chassis: vm
      Machine ID: f8b300b9681541eb9296981c596ee00b
         Boot ID: 47161571fb6c425096a1c3b2db9738f2
  Virtualization: vmware
Operating System: Debian GNU/Linux 12 (bookworm)
          Kernel: Linux 6.1.0-40-amd64
    Architecture: x86-64
 Hardware Vendor: VMware, Inc.
  Hardware Model: VMware Virtual Platform
Firmware Version: 6.00
root@debian:/home/user# nano /etc/hosts
root@debian:/home/user# systemctl restart heartbeat
root@debian:/home/user# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
qlen 1000
    link/ether 00:0c:29:85:1d:47 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 172.20.0.12/24 brd 172.20.0.255 scope global ens33
       valid_lft forever preferred_lft forever
    inet 172.20.0.10/24 brd 172.20.0.255 scope global secondary ens33:0
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe85:1d47/64 scope link
       valid_lft forever preferred_lft forever
root@debian:/home/user#
```

6. Depuis le client Windows : accès à `http://172.20.0.10` -> affichage de **The Grill 2** :



7. Conclusion

Le projet a permis de mettre en place deux serveurs Web fonctionnels et personnalisés, accessibles depuis un client Windows. Grâce à la configuration de Heartbeat, une adresse IP virtuelle assure la continuité de service : en cas de panne du serveur principal, le serveur secondaire prend automatiquement le relais. → **La haute disponibilité est validée et opérationnelle.**